

Gürkan Doğukan Karakurt

GRADUATE RESEARCHER · CIVIL ENGINEER · SOFTWARE DEVELOPER

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MSc candidate and Research Assistant in Construction Engineering & Management at METU, supervised by Prof. Dr. Rifat Sönmez. Thesis applies graph neural networks to BIM models to generate complete construction precedence schedules from a building's geometry and standardized activity classifications. Background in civil engineering (BSc, METU) with a minor in Corporate Finance and a long-running practice in full-stack software development and graphic design.

EDUCATION

- 2025 – Present **MSc, Construction Engineering & Management**
Middle East Technical University (METU), Ankara
Research Assistant. Thesis on BIM-driven scheduling with graph neural networks, supervised by Prof. Dr. Rifat Sönmez. Transferred from the Geotechnical Engineering programme in Spring 2025.
- 2017 – 2023 **BSc, Civil Engineering**
Middle East Technical University (METU), GPA 3.35 / 4.00
Coursework in structural analysis, geotechnics, hydraulics, construction management, and Building Information Modeling.
- 2019 – 2023 **Minor, Corporate Finance**
Middle East Technical University (METU), GPA 3.39 / 4.00
Minor programme covering financial analysis, valuation, and quantitative methods alongside the civil engineering degree.

RESEARCH

Predicting Construction Precedence Relationships from BIM Geometry with Machine Learning

I am building a graph neural network that reads a BIM model and outputs the full construction schedule. The network predicts every FS, SS, FF, SF precedence link, so the schedule regenerates from the model whenever the design changes.

BIM and the construction schedule still do not talk to each other; planners rebuild precedence by hand for every project. The model treats the BIM as a graph: nodes are building elements with their IFC class, geometry, and standardized activity codes (Unifomat II, OmniClass T22); edges encode topology like adjacency, support, and containment. Message passing across that graph scores every candidate precedence link, replacing free-form activity text with standardized vocabularies.

The novel piece is grounding precedence prediction directly in BIM topology and standardized classification codes (IFC, Unifomat II, OmniClass) instead of free-form activity text. That is what lets the model transfer across firms and projects rather than overfit a single dataset.

EXPERIENCE

Oct 2025 – Present	Research Assistant METU · Construction Engineering & Management, Department of Civil Engineering Graduate research and teaching support within the Construction Engineering and Management division. Focus areas include BIM, graph neural networks for construction scheduling, and machine learning applied to the built environment.
Dec 2024 – Oct 2025	Software Engineer Mega Mühendislik, Türkiye Built software for engineering workflows, contributing across the stack to ship reliable internal tooling before moving full-time into graduate research at METU.
2023 – Present	Founder & Full-Stack Developer Diyet Cebimde · AI-assisted diet & nutrition platform Solo-built mobile platform that generates AI-driven weekly diet plans from each user's profile and goals, with photo-based meal recognition, manual tracking, weight and measurement logging, and a memory-equipped chatbot. Hybrid pipeline (Gemini for meal selection, PuLP linear programming for portion optimisation) and snapshot architecture keep historical plans stable across catalog updates. Three repositories: Django + DRF + PostgreSQL backend, React 18 + Vite admin panel, Expo + React Native mobile client.
2023 – 2024	Head of Quantitative Strategies Vortex · Remote (Georgia) Designed and developed algorithmic trading bots for financial markets, including scalping and swing strategies, performance tuning, and per-investor mandate customisation. Owned the strategy roadmap end to end, from market-data ingestion to backtesting to live deployment.
2021 – 2024	Web Development Project Lead (Self-Initiated) Independent · Finance & trading interfaces Built a web-based trading interface for financial markets end to end and ran several smaller finance-side projects in parallel. Picked the stack, owned product decisions, and shipped each iteration to production myself.
Summer 2022	Civil Engineering Intern ES Group, Erdemir Port Project, Zonguldak Worked across site operations, planning, and progress payments on a port construction project. Contributed to steel-pile design and took responsibility in cost control and project scheduling.

SKILLS

Research & Engineering: BIM (Revit, IFC), Construction Management (scheduling, CPM), Machine Learning (PyTorch, GNN (GraphSAGE / GAT)), Optimization (Linear Programming, PuLP), AutoCAD (upper-intermediate), MS Project (intermediate), MATLAB / Visual Basic (advanced)	Backend & Data: Python (advanced), Django + Django REST Framework (advanced), PostgreSQL (ArrayField, JSONField, indexing), JWT (SimpleJWT) + Google OAuth (production auth), Algorithmic trading systems (scalping, swing, custom strategies), Quantitative finance (strategy design, market data, backtesting)
Frontend & Mobile: TypeScript (advanced), React (advanced), Next.js (App Router) (advanced), React Native + Expo (advanced, EAS Build), Tailwind CSS + NativeWind (advanced), React Query + Zustand (state management), Vite, Vercel, Railway (build & deploy), HTML / CSS / JavaScript (advanced)	Languages: Turkish (native), English (C1)

Personal contact details (email, phone, address) are intentionally omitted from this CV for privacy. Please reach out through LinkedIn or ORCID for academic correspondence.